

- 1) A local politician gave a speech and two sentences stood out to you. One was “It’s important that we make contraception (birth control) available and free of cost for everyone, even teenagers.” Another sentence was “Increasing the minimum wage will increase the incomes of poor families in our community.”
- Both sentences are normative claims.
 - The first sentence is a normative claim; the second one is a positive claim.*
 - Both sentences are positive claims.
 - The first sentence is a positive claim; the second one is a normative claim.
 - Most people in the community will probably agree with both those sentences.
- 2) How many widgets would you produce, based on the table below? (Your goal, as always, is to maximize your economic surplus.)

Quantity	Total Cost	Total Benefit	Marginal Benefit
1	2	1	1
2	5	4	3
3	9	9	5
4	13	18	9
5	18	21	3

- 1
 - 2
 - 3
 - 4*
 - 5
- 3) Stacy has recently become interested in purchasing plants for her home. This is a new hobby, and she is debating how many plants she should buy. The total benefit (represented through a monetary amount) of each number of plants is given in the chart below. If the plants cost \$12 each, how many should Stacy purchase?

Number of Plants	Total Benefit (her WTP for that number)
1	20
2	35
3	48
4	57

- 0
 - 1
 - 2
 - 3*
 - 4
- 4) Suppose my willingness to pay for a cookie is higher than your willingness to pay for it. Then we know that
- I will buy more cookies than you.
 - You will buy more cookies than me.

- c) I have a higher income than you.
 - d) [Actually, we can't be sure that any of the other listed options are true.]*
- 5) Suppose you have some money that you use only for investing, and with it you purchased a single bitcoin. After you purchased it, its value fell \$10,000. Then
- a) You should sell your bitcoin now, for fear the price falls further.
 - b) You should hold onto your bitcoin, in hopes that the price jumps back up.
 - c) You should consider your other alternatives; if there is another investment that you think is a better investment than the bitcoin, you should sell your bitcoin and make that other investment.*
 - d) You should trade your bitcoin for another cryptocurrency (another electronic asset) whose price just increased.
- 6) Suppose in September Bobby paid \$200 to reserve a spot for a camping trip in the Shenandoah over Thanksgiving weekend (the \$200 is nonrefundable, but there would be no additional financial expenses associated with the trip). But yesterday an attractive classmate of Bobby's invited Bobby to come to their parents' house in Pittsburgh for Thanksgiving (there would be no need for Bobby to spend any money if Bobby went to Pittsburgh).
- a) If Bobby expects to have more than \$200 worth of enjoyment hanging out with this attractive classmate and their family, Bobby will choose to go to Pittsburgh.
 - b) Bobby compares his current WTP for going camping to his WTP for going to Pittsburgh with the classmate, and he chooses the option for which Bobby has the highest WTP.*
 - c) It depends on what the price of gas is around Thanksgiving.
 - d) Bobby creates his "true WTP" for camping by adding \$200 to his current WTP for going camping. He compares that to his WTP for going to Pittsburgh, and chooses the best option based on that comparison.
- 7) Suppose tomorrow morning you have two final exams. You have studied a lot already, and now you have just three more hours to study. Each hour of studying for your economics exam will increase your score on that exam by an additional 1 point (compared to if you didn't study any more for it). Each hour of studying for your anthropology exam will increase your score on that exam by an additional 2 points. How many hours of economics should you study?
- a) Zero.
 - b) One.
 - c) Two.
 - d) Three.
 - e) Can't say without information about how much you value points on the two exams.*
- 8) Suppose you are the manager of a shop that sells frozen bananas. The owner of the shop (your boss) has insisted that you charge a price of \$3 per banana. Assume at that price there is unlimited demand for your bananas, and that your marginal cost curve is upward sloping. Because your goal is to maximize the shop's economic surplus (i.e., profits), how many bananas do you sell?
- a) As many as you can
 - b) Keep selling more frozen bananas up to the quantity where total revenue (\$3 times the quantity of frozen bananas you sell) is greater than the total cost.
 - c) You keep selling bananas as long as the cost to you of the final banana you sell is not greater than \$3.*
 - d) It depends on the substitutes for frozen bananas that are being sold by other firms in the neighborhood.
- 9) Suppose Bilbo is at an all-you-can-eat sushi buffet. Bilbo already paid \$30 for the buffet, which allowed him to eat as much sushi as he wants. How much sushi did Bilbo eat?

- a) He determined how many pieces of sushi he would be willing to pay \$30 for and ate that many.
 - b) He kept eating sushi until his WTP for the last piece of sushi he ate was 0. Then he stopped.*
 - c) Bilbo ate just enough sushi to satisfy his hunger.
 - d) Bilbo kept eating sushi as long as the MB of an additional piece was greater than 30.
- 10) An individual's demand curve for widgets slopes down because
- a) At a certain point it is difficult to find more widgets at such a low price.
 - b) The MB the individual gets from widgets declines as the individual consumes more widgets.*
 - c) The market supply curve for widgets slopes up.
 - d) The individual gets a negative marginal benefit from an additional widget.
- 11) Suppose Coco is the manager of a widget factory. She is considering hiring additional workers (each worker makes additional widgets), but she has noticed a declining marginal output of labor from the workers. Thus
- a) each additional worker she hires produces fewer widgets than the previous worker.
 - b) she should hire a new worker only if the wage of the worker is less than that worker's marginal revenue product (i.e., the price of widget times the marginal product).
 - c) her supply curve is upward sloping (assuming that the marginal costs for other inputs are constant or increasing)
 - d) [All of the other choices are true.]*
- 12) Under perfect competition, ...
- a) ... some firms make their goods a little bit different to attract more customers.
 - b) ... a firm sells as many units as it wants to at the market price.*
 - c) ... some firms might work together to agree not to lower their prices too much.
 - d) ... a firm's marginal cost curve is not important, because the firm must sell at the market price.
- 13) Rebekah runs a cookie business (this story is familiar from lecture). Each week, at the beginning of the week, before she knows the market price or how many cookies she will sell, she rents a "cookie stand" (fixed cost). In addition to that cost, she also has the variable costs (ingredients, packaging, labor, etc.).
- Suppose last week Rebekah rented a cookie stand for \$10 and she sold 15 batches of cookies. But this week the rental cost of the stand fell to \$5. (Suppose the market price for cookies, and the cost Rebekah faces for other inputs are the same as last week.) Then
- a) Rebekah will sell the same number of cookies.*
 - b) Rebekah will sell more cookies, because now her costs are lower.
 - c) Rebekah will sell fewer cookies, because now her costs are lower.
 - d) Her total profit will be lower.
- 14) Sparrow is planning a BBQ. She goes to the store intending to purchase tofu but instead buys veggie burgers. Suppose that for Sparrow tofu and veggie burgers are substitutes. Then it could have been the case that
- a) the price of tofu was higher than what Sparrow expected.
 - b) the price of veggie burgers was lower than what Sparrow expected.
 - c) the price of some other good that was a complement to veggie burgers and a substitute to tofu fell.
 - d) [All of the other options are valid explanations for Sparrow's behavior.]*

- 15) For Angelica, arepas (a delicious food common in Colombia) are an inferior good (and assume her demand for them is a “downward-sloping, straight line”). When she decided to begin graduate school, her income fell and so her
- a) demand for arepas became more elastic (i.e., the price elasticity of demand became more negative).
 - b) demand for arepas became less elastic.*
 - c) demand was nearly perfectly elastic.
 - d) demand was nearly perfectly inelastic.
- 16) Suppose in England there was a shortage of truck drivers (or lorry drivers, as they call them there). Truck drivers are an important input to gasoline (petrol, as they say), because the drivers deliver gasoline to the various gas stations around the country. What is the effect of the truck driver shortage on the equilibrium price and quantity of gasoline?
- a) The quantity of gasoline falls but the effect on the price is uncertain.
 - b) The quantity of gasoline falls and the price increases.*
 - c) The price of gasoline increases but the effect on the quantity is uncertain.
 - d) The quantity and price of gasoline stays the same because there is no reason supply or demand for gasoline should change.
- 17) The government of Minahstan needs money, so it decided to raise its taxes. But the government knew it would be politically unpopular to announce it was raising taxes on the common people (i.e., the middle and lower class), so instead it implemented a new tax that the grocery stores had to pay. According to our analysis of taxation from class, as a result of the tax increase:
- a) The common people of Minahstan **will not** face higher prices.
 - b) The quantity of goods grocery stores supply **will not** change.
 - c) Most or all of the economic burden of the tax will be borne by grocery stores.
 - d) The common people of Minahstan **will** face higher prices.*
- 18) You are the head barista of a coffee shop whose main product is ... coffee. But the coffee shop also produces a lot of coffee grounds---what is left after making a cup of coffee---which until now never had much value. But a new firm opened up next door that would buy your coffee grounds at a surprisingly high price. Then
- a) You should lower the price of coffee.*
 - b) You should keep the price the same but enjoy higher profits.
 - c) You should raise the price of coffee now that the grounds are worth more.
 - d) You should sell less coffee and focus on producing more grounds.
- 19) David sells very fine boots from his boot shop. He announced that today at 4pm he will begin selling his newest style at a price of \$500/pair. The boots will definitely sell out at that price. Consequently, a line of people starts forming at noon. The price of \$500 and the long line suggests that
- a) Only the people who value the boots the most will get them. (Who else would wait in line for a pair of such expensive boots?)
 - b) Some people who value the boots at \$500 or more will not be able to get a pair.*
 - c) David has priced the boots on the inelastic portion of the demand curve.
 - d) Because of congestion effects, the boots will be worth less than they otherwise might have been.

20) Suppose there is no tax on cigarettes, and they are bought and sold at the equilibrium price. Further suppose that at that price the demand for cigarettes is less elastic than the supply. When the government implements a tax of \$4 dollars a pack on suppliers, then at the new equilibrium

- a) the change in the price buyers pay is larger than the change in the price sellers keep.*
- b) the change in the price buyers pay is smaller than the change in the price sellers keep.
- c) government revenue goes up by \$4 times the quantity sold before there was a tax.
- d) there is no change in the price buyers pay, but sellers have lower profits than before the tax.

21) Suppose the demand function (equation for demand) is $Q_D = 12 - P$ and the supply function (equation for supply is) $Q_S = \frac{1}{2}P$.

- a) The equilibrium quantity is: _____ (4)
- b) The equilibrium price is: _____ (8)
- c) If the price were 2, then there would be a shortage of _____ (9) units
- d) If the government instituted a tax of 3 dollars on sellers, then the new equilibrium quantity would be _____ (3)

22) Sebastian sells chocolate and wanted to figure out the price elasticity of demand of his customers. Almost every week, he charges a price of \$4.50 and sells 105 chocolate bars. But last week he charged a price of \$5.50 and sold 95 bars. Using that data and the midpoint formula, he computed the (arc) price elasticity of demand for his chocolate bars to be:

_____ (-0.5)